

Let us consider the negative-feedback system shown in Fig. 10.1(a), where β is assumed constant. Writing the closed-loop transfer function as

$$\frac{Y}{X}(s) = \frac{H(s)}{1 + \beta H(s)} \quad (10.1)$$

we note that if $\beta H(s = j\omega_1) = -1$ at $\omega_1 \neq 0$, then the closed-loop “gain” goes to infinity, and the circuit can amplify its own noise until it eventually begins to oscillate. In other words, if the loop gain at ω_1 , $\beta H(j\omega_1)$, is equal to -1 , then the circuit may oscillate at frequency ω_1 . This condition can be expressed as

$$|\beta H(j\omega_1)| = 1 \quad (10.2)$$

$$\angle \beta H(j\omega_1) = -180^\circ \quad (10.3)$$

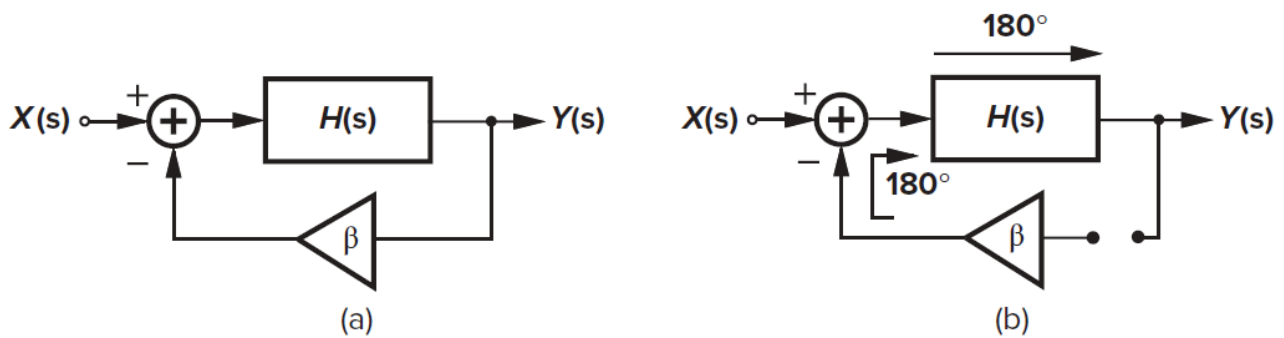


Figure 10.1 (a) Basic negative-feedback system, and (b) phase shift around the loop at ω_1 .

In summary, a negative-feedback system may oscillate at ω_1 if (1) the phase shift around the loop at this frequency is so great that the feedback becomes *positive* and (2) the loop gain is still enough to allow signal buildup. Illustrated in Fig. 10.2(a), the situation can be viewed as excessive loop gain at the frequency at which the phase shift reaches -180° or, equivalently, excessive phase at the frequency at which the loop gain drops to unity. Thus, to avoid instability, we must minimize the total phase shift so that if $|\beta H| = 1$, then $\angle \beta H$ is still more positive than -180° [Fig. 10.2(b)]. In this chapter, we assume that β is less than or equal to unity and does not depend on the frequency.

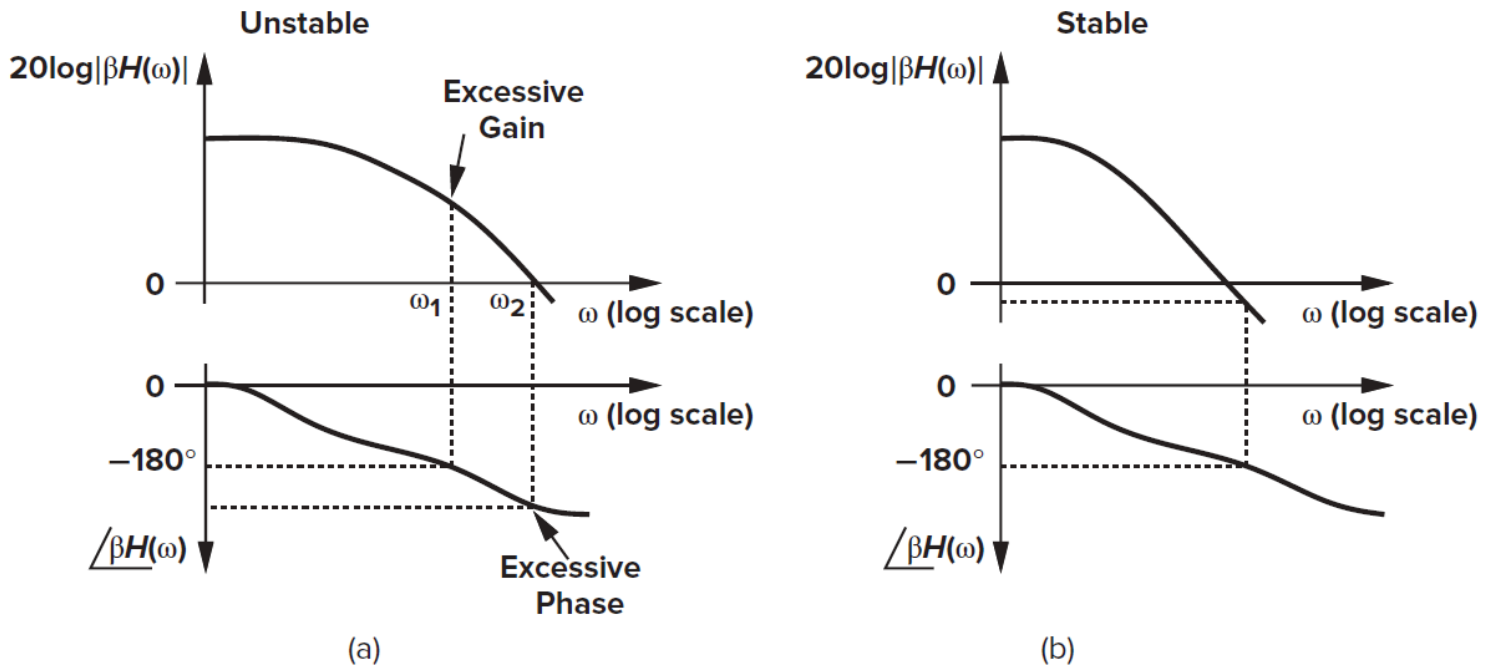


Figure 10.2 Bode plots of loop transmission for (a) unstable and (b) stable systems.

The frequencies at which the magnitude and phase of the loop gain are equal to unity and -180° , respectively, play a critical role in the stability and are called the “gain crossover frequency” and the “phase crossover frequency,” respectively. In a stable system, the gain crossover must occur well before the phase crossover. For the sake of brevity, we denote the gain crossover by GX and the phase crossover by PX. It is helpful to note that the gain crossover frequency is the same as the unity-gain bandwidth of the loop transmission.

- excessive gain and excessive phase
 - if there is less of these excess gain and phase, system is more stable

► Example 10.1

Explain whether the system depicted in Fig. 10.2(a) becomes more or less stable if the feedback is weakened, i.e., if β is reduced.

Solution

As illustrated in Fig. 10.3, a lower β shifts the plot of $20 \log |\beta H(\omega)|$ down and the GX to the left. Since $\angle \beta H(\omega)$ does not change, the system becomes *more* stable. After all, if we apply no feedback around an op amp, the circuit has no tendency to oscillate. Thus, the worst-case stability corresponds to $\beta = 1$, i.e, unity-gain feedback. For this reason, we often analyze the magnitude and phase plots for $\beta H = H$.

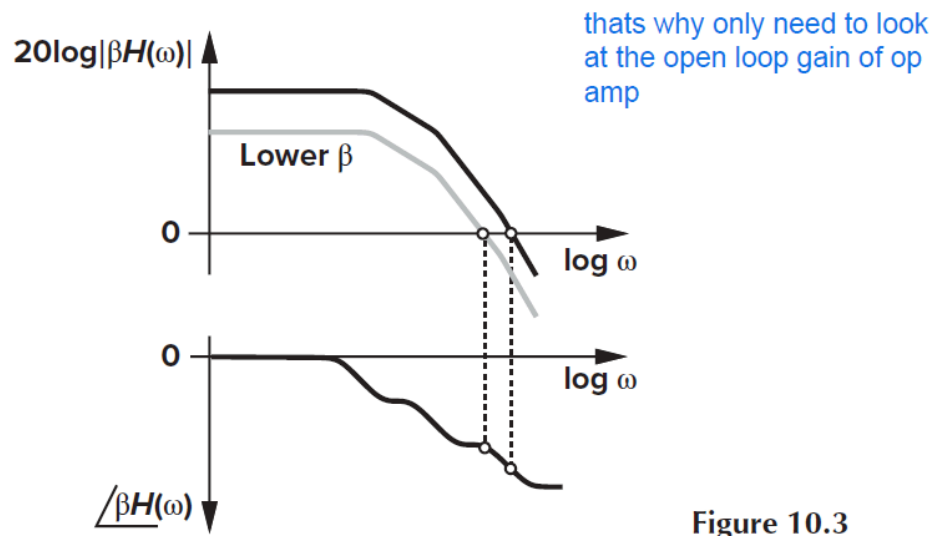


Figure 10.3